

3.5 The Comparison Test & Limit Comparison Test

Motivation: How could we show the series $\sum_{n=1}^{\infty} \left(\frac{1}{1+2^n} \right)$ to be convergent?

One possible way to do this would be by the integral test. However, one way that is perhaps faster is to notice that: $\frac{1}{1+2^n} < \frac{1}{2^n}$ for all $n \geq 1$ and $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right) = 1$. These two facts imply that if $s_k = \sum_{n=1}^k \left(\frac{1}{1+2^n} \right)$ and $t_k = \sum_{n=1}^k \left(\frac{1}{2^n} \right)$, then $s_k < t_k \leq 1$ for all $k \geq 1$.

This means that the sequence of partial sums of $\sum_{n=1}^{\infty} \left(\frac{1}{1+2^n} \right)$ is bounded above by 1. Then, since the sequence of partial sums of $\sum_{n=1}^{\infty} \left(\frac{1}{1+2^n} \right)$ is increasing we know by the Monotonic Sequence Theorem (from section 5.1.2) that the sequence of partial sums of $\sum_{n=1}^{\infty} \left(\frac{1}{1+2^n} \right)$ converges. Hence, $\sum_{n=1}^{\infty} \left(\frac{1}{1+2^n} \right)$ is convergent.

Note: The idea applied above leads to the main theorem of this section which we now state.

Theorem (The Comparison Test): Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with non-negative terms.

1. If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n , then $\sum_{n=1}^{\infty} a_n$ is also convergent.
2. If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n , then $\sum_{n=1}^{\infty} a_n$ is also divergent.

Proof: The proof is very similar to what is mentioned in the motivation section.

Note: When we say “for all n ” in the above theorem we implicitly mean all the possible values that n can take on as an index. The theorem still holds if the index n starts at a value different from one.

Example 1: Determine (with justification) whether the series $\sum_{k=1}^{\infty} \left(\frac{\ln(k)}{k} \right)$ converges or diverges.

Based upon the structure of this series we get the idea to try to compare it to the harmonic series. However, we need to be careful since $\ln(k)$ can be less than one for small values of k .

We note that $\ln(k) > 1$ for $k \geq 3$ (since e is less than 3). This implies that $\frac{\ln(k)}{k} > \frac{1}{k}$ for $k \geq 3$. Since the harmonic series diverges, we know that $\sum_{k=3}^{\infty} \left(\frac{1}{k} \right)$ diverges.

Then, by the Comparison Test, we know that $\sum_{k=1}^{\infty} \left(\frac{\ln(k)}{k} \right)$ is divergent which implies

$$\frac{\ln(1)}{1} + \frac{\ln(2)}{2} + \sum_{k=3}^{\infty} \left(\frac{\ln(k)}{k} \right) = \sum_{k=1}^{\infty} \left(\frac{\ln(k)}{k} \right) \text{ is divergent.}$$

Note: Since we understand the convergence properties of p -series and geometric series quite well, they are good series to consider using when applying the comparison test.

Example 2 (you try): Determine (with justification) whether the series...

(a) $\sum_{n=1}^{\infty} \left(\frac{1}{2n^2 + n} \right)$

We can see that $\frac{1}{2n^2 + n} < \frac{1}{2n^2}$ for $n \geq 1$. We note that $\sum_{n=1}^{\infty} \left(\frac{1}{2n^2} \right) = \frac{1}{2} \sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right)$ which is convergent since

$\sum_{n=1}^{\infty} \left(\frac{1}{n^2} \right)$ is a p -series with $p > 1$. Therefore, by the comparison test, since the series $\sum_{n=1}^{\infty} \left(\frac{1}{2n^2} \right)$ converges the series $\sum_{n=1}^{\infty} \left(\frac{1}{2n^2 + n} \right)$ converges.

(b) $\sum_{n=1}^{\infty} \left(\frac{\arctan(n)}{n^{1.8}} \right)$ converges or diverges.

We recall from Calculus 1 that $0 < \arctan(n) < \frac{\pi}{2}$ for $n \geq 1$. This implies that $0 < \frac{\arctan(n)}{n^{1.8}} < \frac{\pi}{2n^{1.8}}$ for $n \geq 1$.

We note that $\sum_{n=1}^{\infty} \left(\frac{\pi}{2n^{1.8}} \right) = \pi \sum_{n=1}^{\infty} \left(\frac{1}{2n^{1.8}} \right)$. Now, by the p -series test we know that $\sum_{n=1}^{\infty} \left(\frac{1}{n^{1.8}} \right)$ is convergent since

$p = 1.8 > 1$, thus $\sum_{n=1}^{\infty} \left(\frac{\pi}{2n^{1.8}} \right)$ is convergent. Hence, by the Comparison Test, we know that $\sum_{n=1}^{\infty} \left(\frac{\arctan(n)}{n^{1.8}} \right)$ is convergent.

Theorem (The Limit Comparison Test): Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ where c is a positive finite number, then either both series converge or both diverge.

Proof: Suppose $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ where c is a positive, finite number. Then, there are positive numbers m and M such that $m < c < M$.

Now, since $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ we know that there is a positive integer N such that $m < \frac{a_n}{b_n} < M$ when $n > N$. This implies that $mb_n < a_n < Mb_n$ when $n > N$.

Now, if $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} Mb_n$ also converges and $\sum_{n=1}^{\infty} a_n$ converges by the Comparison Test. Also, if $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} mb_n$ also diverges and $\sum_{n=1}^{\infty} a_n$ diverges by the Comparison Test. ■

Note: The above theorem will work if the index starts at a number different from one.

Example 3: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{1}{2^n - 1} \right)$ converges or diverges.

Consider the series $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right)$. Clearly this series and the given series have all positive terms. Now, we see that:

$\lim_{n \rightarrow \infty} \frac{1/(2^n - 1)}{1/2^n} = \lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1} = \lim_{n \rightarrow \infty} \frac{2^n - 1 + 1}{2^n - 1} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{2^n - 1} \right) = 1$. Since this limit exists as a positive finite number and $\sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right)$ is a convergent geometric series, we know by the Limit Comparison Test that the given series is convergent. ■

Note: The solution to this example would have been more difficult to write up if we were set on only using the Comparison Test.

Note: The Limit Comparison Test is inconclusive if we end up getting 0 as our limit.

Example 4 (you try): Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{1}{2n+3} \right)$ converges or diverges.

Consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Clearly this series and the given series have all positive terms. Now,

$$\lim_{n \rightarrow \infty} \frac{1/(2n+3)}{1/n} = \lim_{n \rightarrow \infty} \frac{n}{2n+3} = \lim_{n \rightarrow \infty} \frac{n}{2n+3} \cdot \frac{1/n}{1/n} = \lim_{n \rightarrow \infty} \frac{1}{2-3/n} = \frac{1}{\lim_{n \rightarrow \infty} (2-3/n)} = \frac{1}{2}.$$

Since this limit exists as a positive finite number and the harmonic series is divergent, we know by the Limit Comparison Test that the given series is divergent. ■

We will end this section with a more difficult example.

Example 5: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ converges or diverges.

In this example one might have to experiment with several series to get a feel for which series is the correct series to use for the Comparison Test/Limit Comparison Test. Interestingly, the integral test will not be successful for this series since the function you would need to integrate would be very difficult to integrate.

After some experimentation, one should discover that the harmonic series is the right way to go.

Specifically, consider the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$. Clearly this series and the given series have all positive terms.

Now, we want to know $\lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n}$. For this limit we consider the function $f(x) = \frac{\sin(1/x)}{1/x}$, and we apply

l'Hospital's rule and we find:

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{\sin(1/x)}{1/x} = \lim_{x \rightarrow \infty} \frac{-1/x^2 (\cos(1/x))}{-1/x^2} = \lim_{x \rightarrow \infty} \cos(1/x) = \cos\left(\lim_{x \rightarrow \infty} \frac{1}{x}\right) = \cos(0) = 1.$$

Thus, we have from a theorem from section 9.1 that $\lim_{n \rightarrow \infty} \frac{\sin(1/n)}{1/n} = 1$. Since this limit exists as a positive finite number and the harmonic series is divergent, we know by the Limit Comparison Test that the given series is divergent. ■